Abdelrahman Hesham Ahmed Abdelaziz

Hardware Design Engineer | Power Electronics & Analog PCB | Open to Relocation

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PROFESSIONAL SUMMARY

Hardware design engineer focused on power-electronics and analog PCB design (SMPS, isolated power, Li-ion charging and protection) in Altium Designer. Delivered a 12-unit production run end-to-end, from schematic and layout to client validation. Verifies designs in LTspice and Simulink and debugs boards on the bench. M.Sc. researcher in analog front-end IC design.

WORK EXPERIENCE

Control Systems Engineer | ELKO EP / *Oct 2025 – Sep 2026*

- **PSK Gate System (owned end-to-end):** race-timing gate for athletes at stadium tracks. Prototype, client validation and 12 units delivered, all working; the client has requested further development.
- Designed the schematic and 2-layer PCB (12 V sensor, 5 V internal, 3.3 V Raspberry Pi logic) with optocoupler isolation between the 12 V sensors and the Pi. Two Optex CDD-100N-R sensors per gate feed a logic stage that confirms each crossing.
- Removed sunrise IR interference seen in the client's earlier system by switching to a red-LED sensor. Made sensors and the handheld button detachable over USB Type-C into one enclosure for easy maintenance. Assembled 12 units in 3 weeks.
- **4 automation projects:** two smart-home systems (villas and apartments, design through handover), pump control panels for a water station, and factory control systems. Reviewed approved factory designs and commissioned them on site with the site engineer.

PROJECTS

Dual-Axis Solar Tracker | NTI / *Simulink, Altium Designer*

- Simulink model of the full power chain (PV, CN3791 charger, Li-ion protection, 5 V boost) and tracking control, verified model-in-the-loop against a fixed panel. Designed the 2-layer PCB (CN3791, DW01A/FS8205A, MT3608, INA219, ATmega328P).

Temperature-Controlled DC Fan PCB / *Altium Designer, LTspice*

- Two synchronous buck converters (12 V, 5 V), 3.3 V LDO, STM32F411, TMP36 sensing and a PWM MOSFET fan driver. LTspice buck output ripple of 18.5 mV; BOM and bring-up plan.

CORE-328PB Microcontroller Module / *Altium Designer, Rev A fabricated*

- ATmega328PB with FT232RL USB-UART auto-reset programming and LM1117 LDO with Schottky OR-ed USB/external power. Replaced the castellated reset pad with an on-board manual reset pushbutton.

7-Output Power Supply and Dual-Motor H-Bridge Driver / *Altium Designer*

- Power supply: isolated flyback plus buck stages; fixed +5/+12/+24 V rails (1–2 A) and two adjustable rails (+24 V / −12 V). H-bridge: MOSFET driver for motor direction and speed, 2-layer, fabricated and tested.

Assistive Lower-Limb Exoskeleton / *B.Sc. graduation project, Grade: Excellent*

- Team lead of 6. Fully built, working system: galvanized sheet-metal frame, NEMA 17 motors with custom spur gearboxes, RISC-V controller, gyroscope and limit-switch feedback.

SKILLS

Hardware & PCB: Altium Designer, schematic capture, PCB layout, KiCad, LTspice, SMPS (buck/boost/flyback), isolated power, Li-ion charging, DFM, IPC standards, EMC

Lab & Assembly: Oscilloscope, DMM, SMD soldering, reflow, board bring-up, debugging

Embedded & Tools: C/C++, Python, Git, AVR (ATmega), PIC, STM32, Raspberry Pi, I2C, UART

Modeling & Automation: MATLAB/Simulink, Simscape, Stateflow, PID tuning; Siemens TIA Portal, PLC (Ladder/SCL), SCADA/HMI

TRAINING & CERTIFICATIONS

Electronics Design Engineer Track (SMPS, EMC, DFM) | NTI / *Jul 2026 – Oct 2026*

Model-Based Design Track (BMS capstone) | OS Academy / *May 2026 – Oct 2026*

SCADA Intern | HA Consulting Group / *Sep 2025 – Nov 2025*

Embedded Systems Intern | Information Technology Institute (ITI) / *Aug 2023 – Sep 2023*

Embedded Systems Intern | Professional Company / *May 2023 – Jul 2023*

EDUCATION

M.Sc. Mechatronics Engineering | Ain Shams University, Cairo / *2025 – Nov 2027 (expected)*

- GPA 3.65/4.0. Thesis: Design and Verification of a Precision Analog Front-End IC for Current and Torque Sensing in Robotic Actuator Systems.

B.Sc. Mechatronics Engineering | Higher Technological Institute / *May 2019 – Jan 2024* / *GPA 3.71/4.0 (92.75%, Excellent)*

Languages: Arabic (native), English (B2) **Military status:** Completed